

Datasets & Benchmarks

The corpora and challenges used to train and evaluate detectors

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1 The Codecfake Dataset and Countermeasures for the Universally Detection of Deepfake Audio

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IEEE Trans. Audio, Speech & Language Processing 2025 · arXiv:2405.04880

■ <https://arxiv.org/abs/2405.04880>

ABSTRACT

With the proliferation of Audio Language Model (ALM) based deepfake audio, there is an urgent need for generalized detection methods. ALM-based deepfake audio currently exhibits widespread, high deception, and type versatility, posing a significant challenge to current audio deepfake detection (ADD) models trained solely on vocoded data. To effectively detect ALM-based deepfake audio, we focus on the mechanism of the ALM-based audio generation method, the conversion from neural codec to waveform. We initially constructed the Codecfake dataset, an open-source, large-scale collection comprising over 1 million audio samples in both English and Chinese, focus on ALM-based audio detection. As countermeasure, to achieve universal detection of deepfake audio and tackle domain ascent bias issue of original sharpness aware minimization (SAM), we propose the CSAM strategy to learn a domain balanced and generalized minima. In our experiments, we first demonstrate that ADD model training with the Codecfake dataset can effectively detects ALM-based audio. Furthermore, our proposed generalization countermeasure yields the lowest average equal error rate (EER) of 0.616% across all test conditions compared to baseline models. The dataset and associated code are available online.

2 MLAAD: The Multi-Language Audio Anti-Spoofing Dataset

Nicolas M. Müller, Piotr Kawa, Wei Heng Choong, Edresson Casanova, Eren Gölge, Thorsten Müller, Piotr Syga, Philip Sperl, Konstantin Böttinger

IJCNN 2024 · arXiv:2401.09512

■ <https://arxiv.org/abs/2401.09512>

ABSTRACT

Text-to-Speech (TTS) technology brings significant advantages, such as giving a voice to those with speech impairments, but also enables audio deepfakes and spoofs. The former mislead individuals and may propagate misinformation, while the latter undermine voice biometric security systems. AI-based detection can help to address these challenges by automatically differentiating between genuine and fabricated voice recordings. However, these models are only as good as their training data, which currently is severely limited due to an overwhelming concentration on English and Chinese audio in anti-spoofing databases.

To address this, we present the Multi-Language Audio Anti-Spoof Dataset (MLAAD), created using 52 TTS models, comprising 22 different architectures, to generate 160.2 hours of synthetic voice in 23 different languages. We train and evaluate three state-of-the-art deepfake detection models with MLAAD, and observe that MLAAD demonstrates superior performance over comparable datasets like InTheWild or FakeOrReal when used as a training resource. Furthermore, in comparison with the renowned ASVspoof 2019 dataset, MLAAD proves to be a complementary resource. In tests across eight datasets, MLAAD and ASVspoof 2019 alternately outperformed each other, both excelling on four datasets. By publishing MLAAD and making trained models accessible via an interactive webserver, we aim to democratize antispooofing technology, making it accessible beyond the realm of specialists, thus contributing to global efforts against audio spoofing and deepfakes.

3 ASVspoof 5: Crowdsourced Speech Data, Deepfakes, and Adversarial Attacks at Scale

Xin Wang, Héctor Delgado, Hemlata Tak, Jee-weon Jung, Hye-jin Shim, Massimiliano Todisco, Ivan Kukanov, Xuechen Liu, Md Sahidullah, Tomi Kinnunen, Nicholas Evans, Kong Aik Lee, Junichi Yamagishi

ASVspoof Workshop 2024 · arXiv:2408.08739

■ <https://arxiv.org/abs/2408.08739>

ABSTRACT

ASVspoof 5 is the fifth edition in a series of challenges that promote the study of speech spoofing and deepfake attacks, and the design of detection solutions. Compared to previous challenges, the ASVspoof 5 database is built from crowdsourced data collected from a vastly greater number of speakers in diverse acoustic conditions. Attacks, also crowdsourced, are generated and tested using surrogate detection models, while adversarial attacks are incorporated for the first time. New metrics support the evaluation of spoofing-robust automatic speaker verification (SASV) as well as stand-alone detection solutions, i.e., countermeasures without ASV. We describe the two challenge tracks, the new database, the evaluation metrics, baselines, and the evaluation platform, and present a summary of the results. Attacks significantly compromise the baseline systems, while submissions bring substantial improvements.

4 ADD 2023: the Second Audio Deepfake Detection Challenge

Jiangyan Yi, Jianhua Tao, Ruibo Fu, Xinrui Yan, Chenglong Wang, Tao Wang, Chu Yuan Zhang, Xiaohui Zhang, Yan Zhao, Yong Ren, Le Xu, Junzuo Zhou, Hao Gu, Zhengqi Wen, Shan Liang, Zheng Lian, Shuai Nie, Haizhou Li

IJCAI 2023 Workshop (DADA) · arXiv:2305.13774

■ <https://arxiv.org/abs/2305.13774>

ABSTRACT

Audio deepfake detection is an emerging topic in the artificial intelligence community. The second Audio Deepfake Detection Challenge (ADD 2023) aims to spur researchers around the world to build new innovative technologies that can further accelerate and foster research on detecting and analyzing deepfake speech utterances. Different from previous challenges (e.g. ADD 2022), ADD 2023 focuses on surpassing the constraints of binary real/fake classification, and actually localizing the manipulated intervals in a partially fake speech as well as pinpointing the source responsible for generating any fake audio. Furthermore, ADD 2023 includes more rounds of evaluation for the fake audio game sub-challenge. The ADD 2023 challenge includes three subchallenges: audio fake game (FG), manipulation region location (RL) and deepfake algorithm recognition (AR). This paper

describes the datasets, evaluation metrics, and protocols. Some findings are also reported in audio deepfake detection tasks.

5 Deepfake-Eval-2024: A Multi-Modal In-the-Wild Benchmark of Deepfakes Circulated in 2024

Nuria Alina Chandra et al.

arXiv:2503.02857 (2025)

■ <https://arxiv.org/abs/2503.02857>

ABSTRACT

While many deepfake detectors report high accuracy on academic datasets, we show that these academic benchmarks are out of date and not representative of real-world deepfakes. We introduce Deepfake-Eval-2024, a new deepfake detection benchmark consisting of in-the-wild deepfakes collected from social media and deepfake detection platform users in 2024. Deepfake-Eval-2024 consists of 45 hours of videos, 56.5 hours of audio, and 1,975 images, encompassing the latest manipulation technologies. The benchmark contains diverse media content from 88 different websites in 52 different languages. We find that the performance of open-source state-of-the-art deepfake detection models drops precipitously when evaluated on Deepfake-Eval-2024, with AUC decreasing by 50% for video, 48% for audio, and 45% for image models compared to previous benchmarks. [Abstract excerpt; see the arXiv listing for the complete text.]

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